INCH-POUND

MIL-S-70344
29 June 1990

MILITARY SPECIFICATION

SHIM, LAMINATED,

GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for surface bonded, laminated shims.

1.2 Classification. Laminated shims shall be of the following types and classes, as specified (see 3.1).

Types -
- Type I - All laminations
- Type II - One-half laminations, one-half solid
- Type III - One-fourth laminations, three-fourths solid

Classes -
- Class 1 - .002 inch laminations
- Class 2 - .003 inch laminations

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research, Development and Engineering Center, ATTN:SMCAR-BAC-S, Picatinny Arsenal, NJ 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5365

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:25Z
Check the source to verify that this is the current version before use.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2)

SPECIFICATIONS

FEDERAL

QQ-A-250/8	- Aluminum Alloy 5052, Plate and Sheet
QQ-B-613	- Brass, Leaded and Nonleaded: Flat Products (Plate, Bar, Sheet, and Strip)
QQ-S-766	- Steel Plates, Sheets, and Strip-Corrosion Resisting
PPP-B-636	- Box, Shipping, Fiberboard
PPP-H-1581	- Hardware (Fasteners and Related Items), Packaging of

MILITARY

MIL-S-70344/1	- Shim, Laminated, Round, Class 1 (.002) Laminates
MIL-S-70344/2	- Shim, Laminated, Round, Class 2 (.003) Laminates

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	- Marking for Shipping and Storage
MIL-STD-2073/1	- DOD Materiel Procedures for Development and Application of Packaging Requirements

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Ave., Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/ASME B46.1 - Surface Texture (Surface Roughness, Waviness and Lay)

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018).

Source: https://assist.dla.mil Downloaded: 2026-10-23T03:25Z
Check the source to verify that this is the current version before use.

ASTM

ASTM D3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103).

(Non-government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between requirements of this specification and the specification sheet, the latter shall govern (see 6.2).

3.2 Material. Recycled and reclaimed materials shall be used to the maximum extent practicable. Laminated shims, and partially laminated/ partially solid shims shall be fabricated of the following materials, as specified (see 3.1).

3.2.1 Aluminum alloy. Aluminum alloy shall be Alloy 5052, Temper H38 (UNS A95052) in accordance with QQ-A-250/8.

3.2.2 Brass. Brass shall be Alloy 260, Temper half hard, (UNS C26000) in accordance with QQ-B-613.

3.2.3 Corrosion resistant steel. Corrosion resistant steel shall be Type 302 (UNS S30200), Temper 1/4 hard, in accordance with QQ-S-766.

3.2.4 Bond. The cured adhesive bond shall be within a pH range of 6 to 8. All bonding material shall be compatible with the applicable material specified in 3.2 and meet the construction requirements of 3.3.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:25Z
Check the source to verify that this is the current version before use.

3.3 Construction. The laminated shim stock shall be fabricated in the following three types and two classes, as specified (see 3.1):

Types -
Type I - All laminations
Type II - One-half laminations, one-half solid
Type III - One-fourth laminations, three-fourths solid

Classes -
Class 1 - .002 inch laminations
Class 2 - .003 inch laminations

3.3.1 Construction details. The laminations and solid stock shall be bonded together throughout the whole surface area in a manner which will permit peeling of the laminations for adjustment of the shim thickness without separation of the remaining laminations and solid part. The shim stock shall be capable of being cut, using suitable tools, without separation and shall remain intact with normal handling.

3.3.2 Water immersion. The shim stock shall be capable of withstanding total immersion in water at a temperature of $120^{\circ} \pm 5^{\circ}F$ for a minimum of 3 hours without separation of laminations.

3.4 Dimensions and tolerances. Dimensions and tolerances shall be in accordance with the applicable specification sheet (see 3.1) and as specified herein.

3.4.1 Tolerances.

3.4.1.1 Tolerance on thickness. The tolerance on thickness shall be in accordance with table I.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:25Z
Check the source to verify that this is the current version before use.

TABLE I. TOLERANCES ON THICKNESS

THICKNESS (INCH)	TOLERANCE (PLUS)	TOLERANCE (MINUS)
.006	.0010	.0005
.010	.0015	.0005
.016	.0015	.0010
.020	.0020	.0010
.024	.0020	.0010
.030	.0030	.0020
.032	.0030	.0020
.035	.0030	.0020
.042	.0050	.0020
.048	.0050	.0020
.060	.0060	.0020
.062	.0060	.0020
.063	.0060	.0020
.064	.0060	.0020
.068	.0060	.0020
.072	.0070	.0020
.080	.0070	.0020
.093	.0090	.0030
.094	.0090	.0030
.100	.0100	.0030
.120	.0120	.0030
.125	.0120	.0030
.150	.0150	.0030
.156	.0150	.0030
.405	.0400	.0060

3.4.1.2 Tolerance on lamination thickness. The thickness of Classes 1 and 2 laminations (see 1.2) shall be within ±.0002 inch and ±.0003 inch, respectively, of the values specified herein.

3.4.1.3 Thickness of bond. The thickness of the bond per laminate shall not exceed .0003 inch.

3.5 Surface roughness. Shims shall have a maximum surface roughness of 63 microinches, determined in accordance with ANSI/ASME B46.1.

3.6 Cracks. (See 6.3) Shims shall be free from cracks, flaws and pits in any location.

3.7 Workmanship. Shims shall be free from surface contamination, tool marks and other imperfections which may adversely affect useability.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:25Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Quality conformance inspection. Quality conformance inspection shall be as specified in table II. MIL-STD-105 shall be used as a guide in the development of contractor's statistical techniques to assure the self-locking stud meets all requirements specified herein.

TABLE II. QUALITY CONFORMANCE INSPECTION

INSPECTION	REQUIREMENT PARAGRAPH	TEST PARAGRAPH
Dimensions	3.4	4.3.1
Surface roughness	3.5	4.3.1
Workmanship	3.7	4.3.1

4.2.1 Inspection lot. An inspection lot shall consist of all shims covered by a single specification sheet, of the same material and finish, nominal size, produced by the same manufacturer under essentially the same conditions, and submitted for acceptance at one time. There shall be no evidence of defects in the lot submitted.

4.2.2 Rejected lots. If an inspection lot is rejected, the contractor may rework it to correct the defects, or screen out the defective units and resubmit for inspection. Resubmitted lots shall be inspected in accordance with 4.2. Such lots shall be separate from new lots and shall be clearly identified as reinspected lots.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:25Z
Check the source to verify that this is the current version before use.

4.2.3 Samples for visual and dimensional examination. Samples of shims shall be taken from each lot in accordance with table III and 4.2.

4.2.4 Inspection of packaging. The sampling and inspection of the preservation, packing and container marking shall be in accordance with the requirements of 5.

4.3 Methods of inspection.

4.3.1 Visual and dimensional examination. Samples of shims shall be examined to verify conformance with this specification. Examination shall be conducted in accordance with table III and 4.2.

4.3.1.1 Examination for evidence of cracks, flaws or pits. Samples shall be examined for evidence of cracks, flaws or pits using a 3X or 5X magnification visual aid.

TABLE III. VISUAL AND DIMENSIONAL EXAMINATION

DEFECT	INSPECTION METHOD
Inside diameter, not as specified (see 3.4)	CIE 1/
Outside diameter, not as specified (see 3.4)	CIE 1/
Thickness, not as specified (see 3.4)	CIE 1/
Evidence of cracks, flaws or pits (see 3.6)	Visual
Surface roughness, not as specified (see 3.5)	CIE 1/
Workmanship (see 3.7)	Visual

1/ Commercial inspection equipment

4.3.2 Material inspection. Material inspection shall consist of certification supported by verifying data that the materials used in fabricating the shims are in accordance with the applicable paragraphs listed in 3.2, 3.3, and 3.4 prior to such fabrication.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:25Z
Check the source to verify that this is the current version before use.

5. PACKAGING

5.1 Preservation. Preservation shall be Level A, or Commercial as specified (see 6.2).

a. Level A shall be in accordance with PP-H-1581.

b. Commercial shall be in accordance with ASTM D3951.

5.2 Packing. Packing shall be Level A, B, C or Commercial as specified (see 6.2).

a. Level A shall be in accordance with PPP-H-1581 except PPP-B-636 weather resistant shall not be used as a Level A shipper.

b. Level B shall be in accordance with PPP-H-1581 except PPP-B-636 weather resistant may be used for Level B overseas shipment.

c. Level C shall be in accordance with MIL-STD-2073-1.

d. Commercial shall be in accordance with ASTM D3951.

5.3 Marking. Marking shall be Level A, B and Commercial. In addition to any special or other identification marking required by the contract (see 6.2) each unit pack intermediate and exterior container and unitized load shall be marked in accordance with MIL-STD-129/ASTM D3951, as applicable.

5.4 Exterior container. Exterior containers (see 5.2 a, b, c and d) shall be of minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent possibie.

6. NOTES

6.1 Intended use. Shims acquired under this specification are intended to provide adjustment of fits and alignment of component parts of an assembly by peeling of one or more laminations.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification and applicable specification sheet.

b. Applicable specification sheet part number (see 3.1).

c. Selection of applicable levels of preservation, packing, marking and commercial/industrial (see 5.1, 5.2 and 5.3).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:25Z
Check the source to verify that this is the current version before use.

6.3 Definition. A crack is a clean crystalline fracture passing through or across the grain boundaries without inclusion of foreign elements.

6.4 Subject term (key word) listing.

Spacer
Washer, Flat
Spacer, Ring

Custodians:
Army - AR
Air Force - 99

Review activities:
Army - AV, MI
Air Force - 82
DLA - IS

User activities:
Army - AT
NSA - NS

Preparing activity:
Army - AR

Agent:
DLA - IS

(Project 5365-0121)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:25Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8 In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-S-70344	29 June 1990 .

3. DOCUMENT TITLE

SHIM, LAMINATED, GENERAL SPECIFICATION FOR -

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible Attach extra sheets as needed)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b TELEPHONE (Include Area Code)	
US Army ARDEC Standardization & Specification Office	(1) Commercial (201) 724-6628	(2) AUTOVON 880-6628
ADDRESS (Include Zip Code) ATTN: SMCAR-BAC-S Picatinny Arsenal, NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:25Z
Check the source to verify that this is the current version before use.